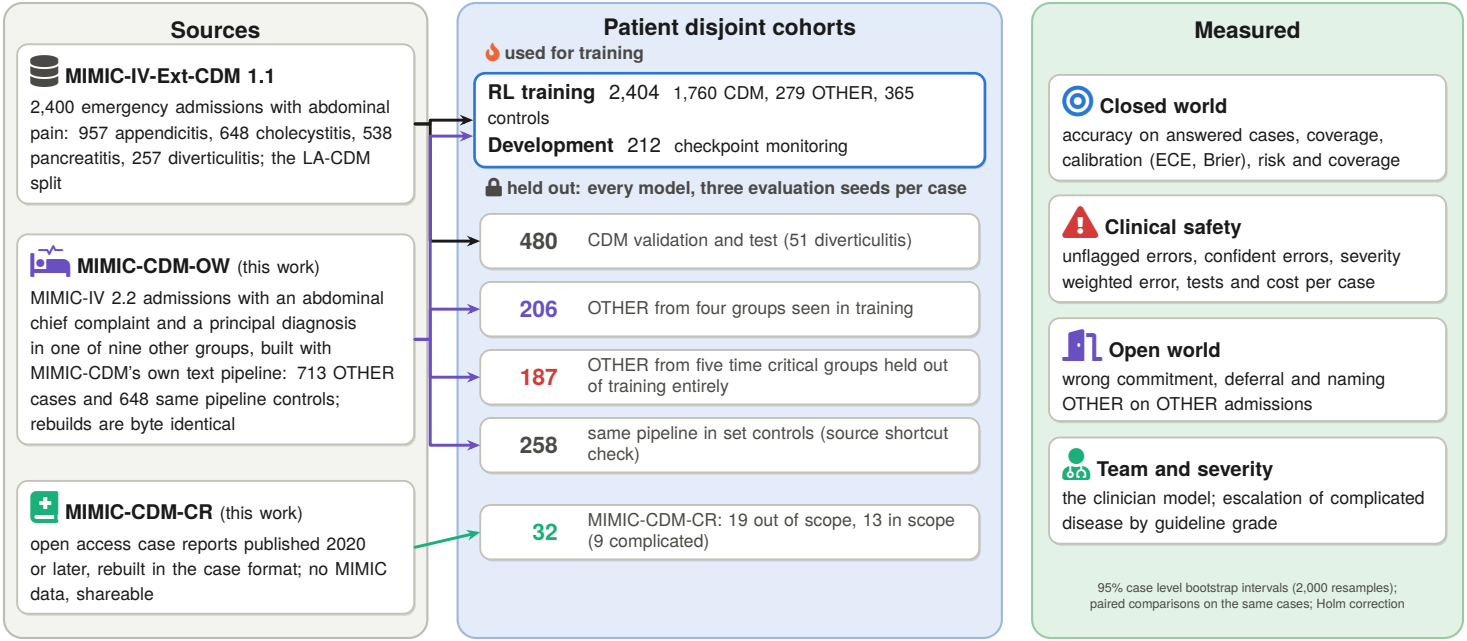
Sources
MIMIC-IV-Ext-CDM 1.1
2,400 emergency admissions with abdominal pain: 957 appendicitis, 648 cholecystitis, 538 pancreatitis, 257 diverticulitis; the LA-CDM split
MIMIC-CDM-OW (this work)
MIMIC-IV 2.2 admissions with an abdominal chief complaint and a principal diagnosis in one of nine other groups, built with MIMIC-CDM's own text pipeline: 713 OTHER cases and 648 same pipeline controls; rebuilds are byte identical
MIMIC-CDM-CR (this work)
open access case reports published 2020 or later, rebuilt in the case format; no MIMIC data, shareable
Patient disjoint cohorts
used for training
RL training 2,404 1,760 CDM, 279 OTHER, 365 controls
Development 212 checkpoint monitoring
held out: every model, three evaluation seeds per case
480 CDM validation and test (51 diverticulitis)
206 OTHER from four groups seen in training
187 OTHER from five time critical groups held out of training entirely
258 same pipeline in set controls (source shortcut check)
32 MIMIC-CDM-CR: 19 out of scope, 13 in scope (9 complicated)
Measured
Closed world
accuracy on answered cases, coverage, calibration (ECE, Brier), risk and coverage
Clinical safety
unflagged errors, confident errors, severity weighted error, tests and cost per case
Open world
wrong commitment, deferral and naming OTHER on OTHER admissions
Team and severity
the clinician model; escalation of complicated disease by guideline grade
95% case level bootstrap intervals (2,000 resamples); paired comparisons on the same cases; Holm correction